Substance use disorders

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Discuss the use, disorder, intoxication, dependence and withdrawal of the commonly encountered substances in the clinical setting

Explore resources available for providers and patients in need of information regarding substance use disorders

resources

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- https://www.ncbi.nlm.nih.gov/books/NBK565474/table/nycgsubuse.tab9/#_ncbi_dlg_citbx_NBK565474
- https://www.ncbi.nlm.nih.gov/books/NBK557381/#_ncbi_dlg_citbx_NBK557381
- Jesse S, Bråthen G, Ferrara M, Keindl M, Ben-Menachem E, Tanasescu R, Brodtkorb E, Hillbom M, Leone MA, Ludolph AC. Alcohol withdrawal syndrome: mechanisms, manifestations, and management. Acta Neurol Scand. 2017 Jan;135(1):4-16. doi: 10.1111/ane.12671. Epub 2016 Sep 1. PMID: 27586815; PMCID: PMC6084325.
- https://www.niaaa.nih.gov
- Ziedonis D, Das S, Larkin C. Tobacco use disorder and treatment: new challenges and opportunities. Dialogues Clin Neurosci. 2017 Sep;19(3):271-280. doi: 10.31887/DCNS.2017.19.3/dziedonis. PMID: 29302224; PMCID: PMC5741110.
- McLaughlin I, Dani JA, De Biasi M. Nicotine withdrawal. Curr Top Behav Neurosci. 2015;24:99-123. doi: 10.1007/978-3-319-13482-6_4. PMID: 25638335; PMCID: PMC4542051.
- Patel J, Marwaha R. Cannabis Use Disorder. [Updated 2022 Jul 11]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK538131/
- DSM 5 pages 490-585

Substance use disorder (SUD) is a complex condition in which there is uncontrolled use of a substance despite harmful consequences. People with SUD have an intense focus on using a certain substance(s) such as alcohol, tobacco, or illicit drugs, to the point where the person's ability to function in day-to-day life becomes impaired. People keep using the substance even when they know it is causing or will cause more harm than good.

SUDs are classified as mild, moderate, or severe based on how many of the 11 criteria are fulfilled: mild, any 2 or 3 criteria; moderate, any 4 or 5 criteria; severe, any 6 or more criteria.

Elevated risk of suicide

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Impaired control over substance use (DSM-5 criteria 1 to 4)	Consuming the substance in larger amounts and for a longer amount of time than intended. •Persistent desire to cut down or regulate use. The individual may have unsuccessfully attempted to stop in the past. •Spending a great deal of time obtaining, using, or recovering from the effects of substance use. •Experiencing craving, a pressing desire to use the substance.

Social impairment (*DSM-5* criteria 5 to 7)	•Substance use impairs ability to fulfill major obligations at work, school, or home. •Continued use of the substance despite it causing significant social or interpersonal problems. •Reduction or discontinuation of recreational, social, or occupational activities because of substance use.

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RISKY USE (*DSM-5* CRITERIA 8 AND 9)

- RECURRENT SUBSTANCE USE IN PHYSICALLY UNSAFE ENVIRONMENTS.
- PERSISTENT SUBSTANCE USE DESPITE KNOWLEDGE THAT IT MAY CAUSE OR EXACERBATE PHYSICAL OR PSYCHOLOGICAL PROBLEMS.

Pharmacologic (DSM-5 criteria 10 and 11)	•Tolerance: Individual requires increasingly higher doses of the substance to achieve the desired effect, or the usual dose has a reduced effect; individuals may build tolerance to specific symptoms at different rates. •Withdrawal: A collection of signs and symptoms that occurs when blood and tissue levels of the substance decrease. Individuals are likely to seek the substance to relieve symptoms. No documented withdrawal symptoms from hallucinogens, PCP, or inhalants. •Note: Individuals can have an SUD with prescription medications, so tolerance and withdrawal (criteria 10 and 11) in the context of appropriate medical treatment do not count as criteria for an SUD.

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Binge drinking: BAC to 0.08%, women>/=4 drinks, men>/=5drinks within 2 hours

Heavy drinking: men>/=5 a day/15 per week, women >/=4 a day or >/=8 per week

High intensity drinking: two times binge drinking thresholds

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84% of people over 18 yrs old have had a drink/23% report binging

23% of youths aged12-17 reported alcohol use/3% report binge drinking

0.4% of 12 to 17 yr olds reported heavy alcohol consumption

6.4% of 18yr olds and older reported heavy alcohol consumption

Greater than 140,000 people died from alcohol related causes 2015-2019

Alcohol contributed 7% of ED visits and 17% of opioid overdoses

A chronic, relapsing brain disorder characterized by an impaired ability to stop or control alcohol use despite adverse social, occupational, or health consequences

Ranges from mild to severe

Recovery is always possible

Includes alcohol abuse and alcohol dependence

3.4% of 12 to 17 yr olds

11.3 % of 18 yrs and older

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In 2010, alcohol misuse cost the United States $249 billion.[1]

Three quarters of the total cost of alcohol misuse is related to binge drinking.[1]

Reference

Sacks JJ, Gonzales KR, Bouchery EE, Tomedi LE, Brewer RD. 2010 national and state costs of excessive alcohol consumption. Am J Prev Med. 2015;49(5):e73-e79. PubMed PMID: 26477807

Alcohol	Unsteady gait, slurred speech, nystagmus, flushed face, conjunctival injection, decreased levels of consciousness	Disinhibition, argumentativeness, aggression, inattention, lability of mood, impaired judgement and functioning

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Alcohol poisoning can result from drinking any type of alcohol, including beer, wine or liquor. As your stomach digests and absorbs alcohol, the alcohol enters your bloodstream, and your alcohol blood level begins to rise. Your liver breaks down alcohol. But when blood alcohol levels are high, your overwhelmed liver can’t remove the toxins quickly enough.

The extra alcohol in the bloodstream is a depressant. That means it reduces normal function. In this case, it affects the parts of the brain that control vital body functions, such as breathing, heart rate, blood pressure and temperature. As blood alcohol continues to rise, the depressant effect is more substantial

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- Bluish-colored or cold, clammy skin, especially around the lips and fingernails.
- Confusion, slowed responses, lack of coordination or being unable to walk.
- Difficulty remaining conscious.
- Hypothermia.
- Irregular pulse, heartbeat or breathing (intervals of 10 seconds or more between breaths).
- Problems with bladder or bowel control (incontinence).
- Seizures, vomiting or choking.
- Strong alcohol odor

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Breathalyzer: As you drink, the alcohol goes through your bloodstream to your lungs. There, it evaporates into the lungs, and you breathe it out. As you blow into the breathalyzer, it can estimate your BAC by how much alcohol it detects in your breath.

Blood test: A lab technician draws a small amount of blood with a needle. The technician then analyzes for the BAC. The blood test is most accurate within six to 12 hours after the last drink you consume.

Management of Alcohol Poisoning

- Maintain airway
- IV fluids: intravenous (IV) fluids to treat dehydration.
- Glucose: many patients are hypoglycemic
- Thiamine: prevention of Wernickes encephalitis
- Haloperidol: If patient is agitated
- Full metabolic panel and liver enzymes, ECG,
- maybe CT if altered mental status and no hx available

Alcohol withdrawal syndrome

May occur within hours, peaks at 72 hours, results in the absence of GABAergic input of alcohol

May include tremors, insomnia, agitation, diaphoresis, hypertension, tachycardia or seizures

Hallucinations may occur and signal alcohol withdrawal DELERIUM (2% of patients)

Any history of previous withdrawal from alcohol that includes delirium or seizures should be treated in a monitored setting

Alcohol Use Disorder Brick

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https://exchange.scholarrx.com/brick/1z050x1oy5dv

Tobacco use disorder

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- **Main effects of neurotransmitters that are released by nicotine binding. GABA, γ-aminobutyric acid**

- Adapted from reference 18: Benowitz. Clinical pharmacology of nicotine: implications for understanding, preventing, and treating tobacco addiction. *Clin Pharmacol Ther.* 2008;83(4):531-541.

Neurotransmitter	Effect
Dopamine	Pleasure, appetite suppression
Norepinephrine	Arousal, appetite suppression
Acetylcholine	Arousal, cognitive enhancement
Glutamate	Learning, memory enhancement
Serotonin	Mood modulation, appetite suppression
β-Endorphin	Reduction in anxiety and tension
GABA	Reduction in anxiety and tension

Nicotine withdrawal

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- The Diagnostic and Statistical Manual of Mental Disorders (DSM-5) reports 7 primary symptoms associated with nicotine withdrawal: irritability/anger/frustration, anxiety, depressed mood, difficulty concentrating, increased appetite, insomnia, and restlessness
- There are seven FDA-approved medications for smoking cessation, including two non-nicotine replacement pill options of bupropion (Zyban) and varenicline (Chantix) and five nicotine-replacement therapies (NRTs).
- Varenicline (Chantix) appears to be the most effective compared with placebo, followed by bupropion and NRTs. Varenicline is a partial agonist at the α4β2 receptors; as a partial agonist, it relieves craving and withdrawal, but reduces the reinforcing effects of nicotine by blocking dopaminergic stimulation (less smoking reinforcement/reward).
- Bupropion (Zyban) is the other medication that can affect dopamine, norepinephrine, and nicotinic-cholinergic receptors to decrease cravings and withdrawal symptoms.
- NRTs function to satisfy nicotine receptors while being less reinforcing.

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- https://exchange.scholarrx.com/brick/tobacco-smoking

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Cannabis use disorder

- The acute phase includes intoxication and withdrawal states, along with secondary complications - delirium, psychosis, anxiety, and insomnia. Chronic regular use can be characterized by disordered behavior (DMS 5)

Recent use of cannabis

↓

Clinically significant problematic behavioral or psychological changes (e.g., impaired motor coordination, euphoria, anxiety, a sensation of slowed time, impaired judgment, social withdrawal) that developed during, or shortly after, cannabis use

↓

At least two of the following signs, developing within 2 hours of cannabis use: conjunctival injection, increased appetite, dry mouth, tachycardia.

↓

Symptoms not due to a general medical condition and are not better accounted for by another mental disorder.

↓

Specify if perceptual disturbances are present: hallucinations with intact reality testing or auditory, visual, or tactile illusions occur in the absence of delirium.

This accompanies cessation of cannabis use that has been heavy and prolonged (i.e., usually daily or almost daily use over a period of at least a few months). Three or more of the following signs and symptoms develop within approximately 1 week after cessation of heavy, prolonged use:

- Irritability, anger, or aggression
- Nervousness or anxiety
- Sleep difficulty (i.e., insomnia, disturbing dreams)
- Decreased appetite or weight loss
- Restlessness
- Depressed mood

At least one of the following physical symptoms causing significant discomfort: abdominal pain, shakiness/tremors, sweating, fever, chills, or a headache.

The signs or symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

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- https://exchange.scholarrx.com/brick/cannabis-use-disorder

Opioid use disorder

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natural: from the poppy plant/morphine & codeine

semi synthetic: heroine, oxycodone, hydrocodone

synthetic: fentanyl, methadone, tramadol

- 0.37% of adults 18 yrs and older
- Usually starts in late teens/early twenties
- Genetic factors play a particularly important role
- At risk for TB, HIV, endocarditis, cellulitis, hepatitis

- Fentanyl is 50 times more potent than heroin and 100 times more potent than morphine

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Recent use of an opioid.

B. Clinically significant problematic behavioral or psychological changes (e.g., initial euphoria followed by apathy, dysphoria, psychomotor agitation or retardation, impaired judgment) that developed during, or shortly after, opioid use.

C. Pupillary constriction (or pupillary dilation due to anoxia from severe overdose) and one (or more) of the following signs or symptoms developing during, or shortly after, opioid use: 1. Drowsiness or coma. 2. Slurred speech. 3. Impairment in attention or memory.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

- M:miosis
- O:out of it (sedation)
- R: respiratory depression
- P:pneumonia (aspiration)
- H: hypotension
- I: infrequency (constipation and urinary retention)
- N: nausea
- E: emesis

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A. Presence of either of the following: 1. Cessation of (or reduction in) opioid use that has been heavy and prolonged (i.e., several weeks or longer). 2. Administration of an opioid antagonist after a period of opioid use.

B. Three (or more) of the following developing within minutes to several days after Criterion A: 1. Dysphoric mood. Nausea or vomiting. Muscle aches. Lacrimation or rhinorrhea. Pupillary dilation, piloerection, or sweating. Diarrhea. Yawning. Fever. Insomnia.

C. The signs or symptoms in Criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Treatment of Opioid Use Disorder

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Methadone is an opioid agonist/eliminates withdrawal and reduces cravings

Buprenorphine is a partial opioid agonist/reduces withdrawal and cravings

Naltrexone is an opioid antagonist/prevent opioids from producing rewarding effects

Methadone=Buprenorphine>naltrexone

All are more effective with counseling/behavioral therapy

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Opioid Use Disorder brick

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Hallucinogens are a class of drugs that cause hallucinations—profound distortions in a person's perceptions of reality. Hallucinogens can be found in some plants and mushrooms (or their extracts) or can be man-made, and they are commonly divided into two broad categories: classic hallucinogens (such as LSD) and dissociative drugs (such as PCP). When under the influence of either type of drug, people often report experiencing rapid, intense emotional swings and seeing images, hearing sounds, and feeling sensations that seem real but are not.

PCP-PHENCYCLIDINE	KETAMINE	MESCALINE
LSD-D-4LYSERIC ACID DIETHYLAMIDE	PSILOCYBIN	BATH SALTS
DXM-DEXTROMETHORPHAN	SALVIA DIVINORUM	ECSTASY

A. Recent use of phencyclidine (or a pharmacologically similar substance).

B. Clinically significant problematic behavioral changes (e.g., belligerence, assaultiveness, impulsiveness, unpredictability, psychomotor agitation, impaired judgment) that developed during, or shortly after, phencyclidine use.

C. Within 1 hour, two (or more) of the following signs or symptoms: Note: When the drug is smoked, "snorted," or used intravenously, the onset may be particularly rapid. 1. Vertical or horizontal nystagmus. 2. Hypertension or tachycardia. Numbness or diminished responsiveness to pain. Ataxia. Dysarthria. Muscle rigidity. Seizures or coma. Hyperacusis.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance

Recent intended or unintended short-term, high-dose exposure to inhalant substances, including volatile hydrocarbons such as toluene or gasoline.

B. Clinically significant problematic behavioral or psychological changes (e.g., belligerence, assaultiveness, apathy, impaired judgment) that developed during, or shortly after, exposure to inhalants.

C. Two (or more) of the following signs or symptoms developing during, or shortly after, inhalant use or exposure: Dizziness. Nystagmus. Incoordination. Slurred speech. Unsteady gait. Lethargy. Depressed reflexes. Psychomotor retardation. Tremor. Generalized muscle weakness. Blurred vision or diplopia. Stupor or coma. Euphoria.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

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A. Recent use of a sedative, hypnotic, or anxiolytic.

B. Clinically significant maladaptive behavioral or psychological changes (e.g., inappropriate sexual or aggressive behavior, mood lability, impaired judgment) that developed during, or shortly after, sedative, hypnotic, or anxiolytic use.

C. One (or more) of the following signs or symptoms developing during, or shortly after, sedative, hypnotic, or anxiolytic use: Slurred speech. Incoordination. Unsteady gait. Nystagmus. Impairment in cognition (e.g., attention, memory). Stupor or coma.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

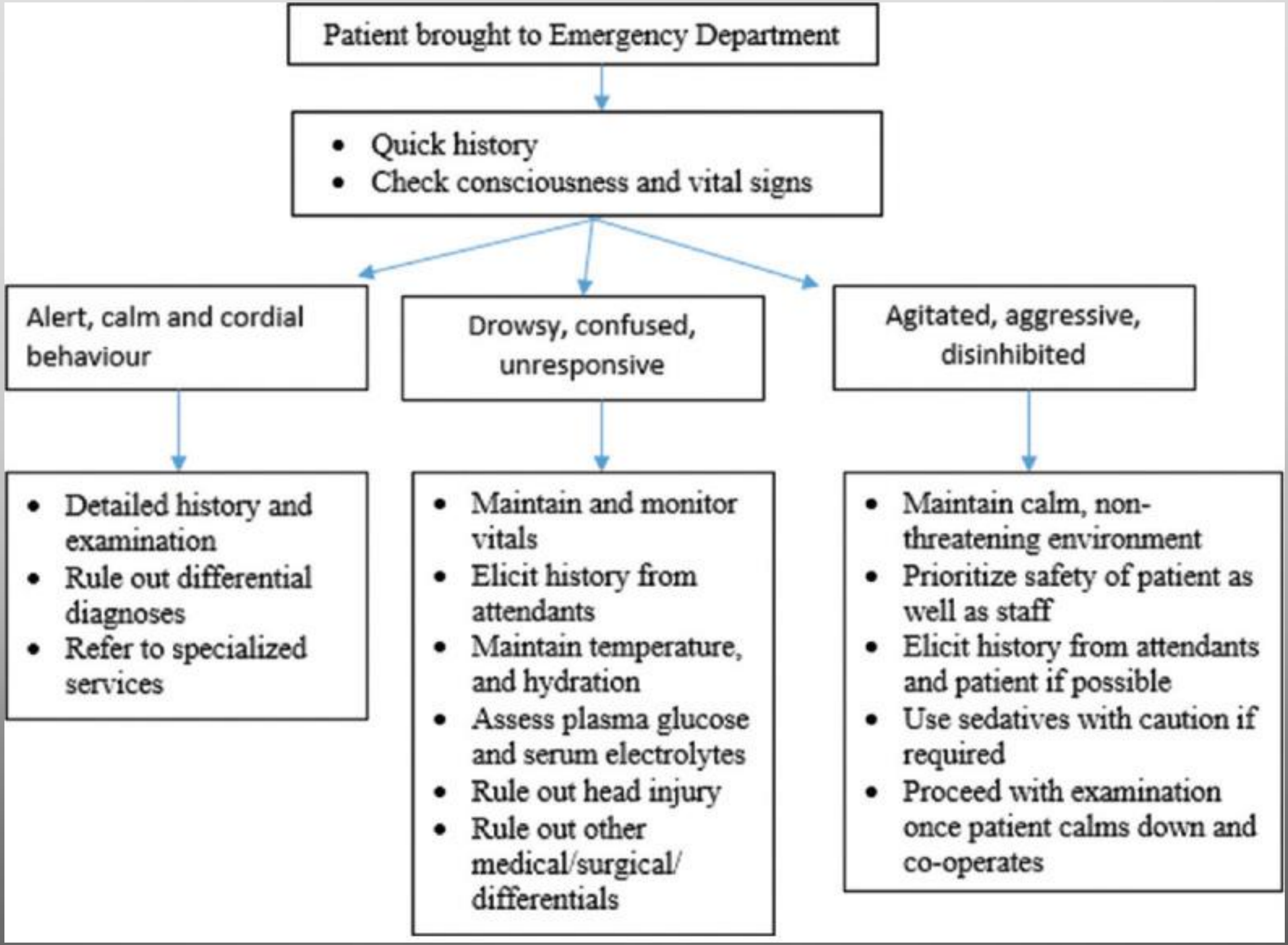
Differentiate from alcohol intoxication

A. Cessation of (or reduction in) sedative, hypnotic, or anxiolytic use that has been prolonged.

B. Two (or more) of the following, developing within several hours to a few days after the cessation of (or reduction in) sedative, hypnotic, or anxiolytic use described in Criterion A: 1. Autonomic hyperactivity (e.g., sweating or pulse rate greater than 100 bpm). 2. Hand tremor. Insomnia. . Nausea or vomiting. . Transient visual, tactile, or auditory hallucinations or illusions. . Psychomotor agitation. Anxiety. . Grand mal seizures.

C. The signs or symptoms in Criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.


Patient brought to Emergency Department
Quick history
Check consciousness and vital signs
Alert, calm and cordial behaviour
Drowsy, confused, unresponsive
Agitated, aggressive, disinhibited
Detailed history and examination
Rule out differential diagnoses
Refer to specialized services
Maintain and monitor vitals
Elicit history from attendants
Maintain temperature, and hydration
Assess plasma glucose and serum electrolytes
Rule out head injury
Rule out other medical/surgical/ differentials
Maintain calm, non-threatening environment
Prioritize safety of patient as well as staff
Elicit history from attendants and patient if possible
Use sedatives with caution if required
Proceed with examination once patient calms down and co-operates

A. Recent use of an amphetamine-type substance, cocaine, or other stimulant.

B. Clinically significant problematic behavioral or psychological changes (e.g., euphoria or affective blunting; changes in sociability; hypervigilance; interpersonal sensitivity; anxiety, tension, or anger; stereotyped behaviors; impaired judgment) that developed during, or shortly after, use of a stimulant.

C. Two (or more) of the following signs or symptoms, developing during, or shortly after, stimulant use: Tachycardia or bradycardia. Pupillary dilation. Elevated or lowered blood pressure. Perspiration or chills. Nausea or vomiting. Evidence of weight loss. Psychomotor agitation or retardation. Muscular weakness, respiratory depression, chest pain, or cardiac arrhythmias. Confusion, seizures, dyskinesias, dystonias, or coma.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Chest pain, palpitations are a frequent presenting symptom.

A. Cessation of (or reduction in) prolonged amphetamine-type substance, cocaine, or other stimulant use.

B. Dysphoric mood and two (or more) of the following physiological changes, developing within a few hours to several days after Criterion A: Fatigue. Vivid, unpleasant dreams. Insomnia or hypersomnia. Increased appetite. Psychomotor retardation or agitation.

C. The signs or symptoms in Criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

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Resources for providers and patients

- SAMHSA’s National Helpline at **1-800-662-HELP (substance abuse and mental health services admin)**
- **https://findtreatment.gov/**
- https://www.cdc.gov/
- https://www.cdc.gov/drugoverdose/
- Oasas.ny.gov for naloxone samples and training
- Summary of the 2022 Clinical Practice Guideline for Prescribing Opioids for Pain
- PMP – mandatory checking before prescribing

Lecture and Lab Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R